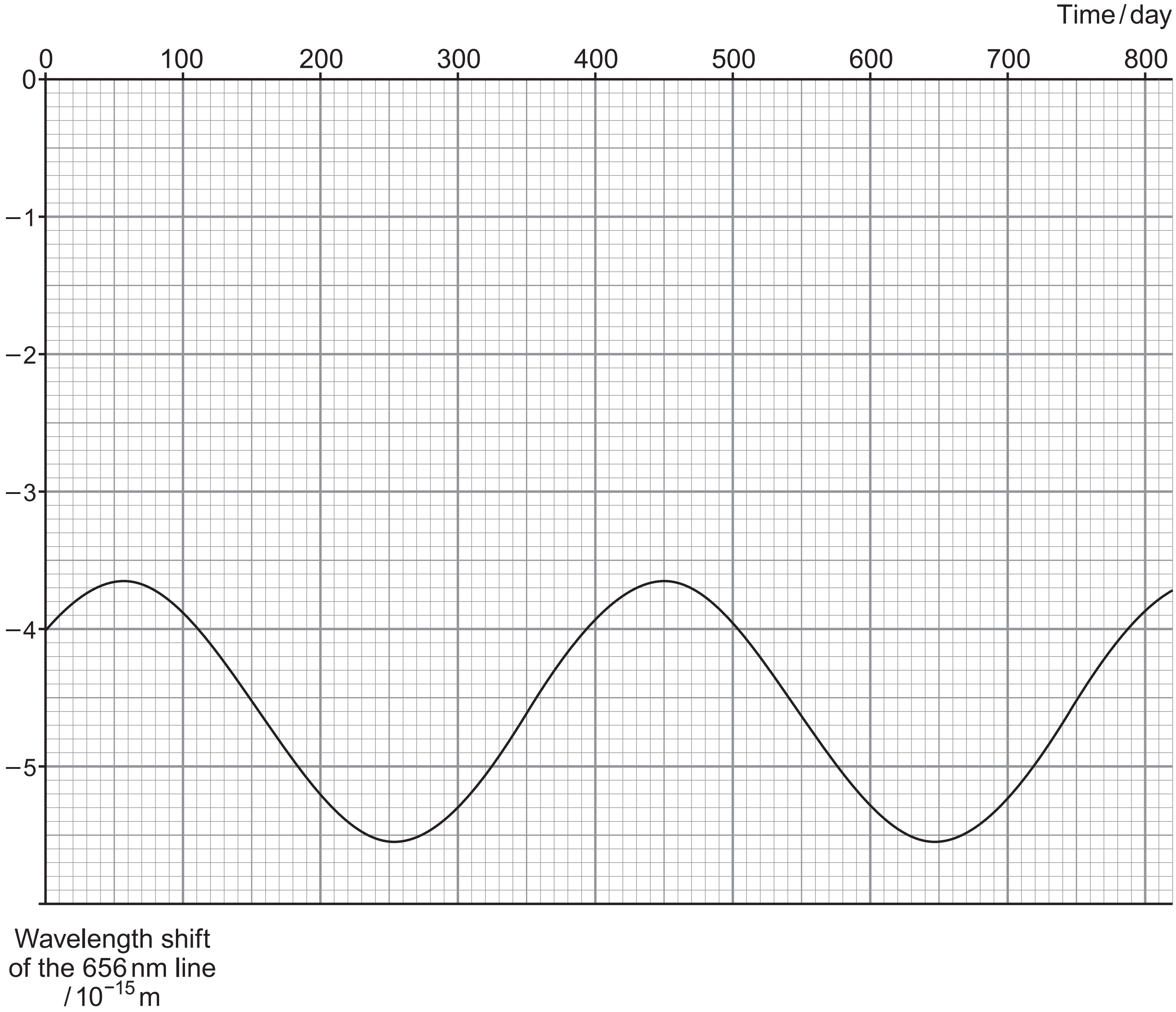
3. In the future, analysis of the 656 nm red hydrogen line from a distant star might reveal the following Doppler variation in wavelength (the star would be viewed edge-on).



The mean wavelength is shorter than 656 nm on average because of the mean radial velocity of the star but the sinusoidal variation is caused by an orbiting planet.

Examiner only

(a) Use the data in the graph to show that the mean radial velocity of the star is approximately $-2\,\mathrm{m\,s^{-1}}$ and that the orbital speed of the star around the centre of mass is approximately $0.4\,\mathrm{m\,s^{-1}}$. [4]

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(b) Explain briefly whether the light is blue-shifted or red-shifted. [1]

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(c) Show that the radius of the star's orbit is approximately 2 000 km. [3]

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Examiner only

(d) The star has a mass of 2.07×10^{30} kg.

(i) Calculate the distance between the star and the planet. [3]

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(ii) Calculate the mass of the orbiting planet (assume that the mass of the planet is much smaller than the mass of the star). [2]

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(e) The values of part (d) reveal that the star is very similar in mass to the Sun, the planet's distance from the star is very close to the distance between the Sun and Earth but that the mass of the planet is approximately 5 times that of the Earth. Discuss whether or not it is scientifically reasonable to conclude that life exists on this planet. [3]

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